Google Cloud H2S HACKATHONS

AI Builder Cup

Build in JAPAC.
Build for the world.



Ada-Twin

The adaptive factory twin.

Turn a single 360° photo of a plant floor into a walkable 3D digital twin, then let AI agents run a simulated shift on it to find the bottleneck and test fixes before spending money on the real floor.

TEAM

[Team name]

TEAM LEADER

[Leader name]

MEMBERS

[Members]

PROBLEM STATEMENT

Manufacturing: Intelligent Operations & Industrial Efficiency

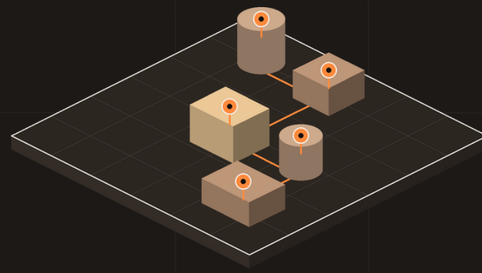
01 — BRIEF ABOUT THE IDEA

Ada-Twin :Your plant floor, as a living simulation

A digital twin normally needs CAD models, laser scans and sensors. So small and mid-sized plants decide layouts, staffing and maintenance by gut feel.



ONE 360° PHOTO



WALKABLE 3D TWIN

01 · RECONSTRUCT

Photo to walkable 3D

One 360° panorama becomes a 3D world with real-world scale.

~12 min

to generate the twin

›

02 · DISCOVER

Find every machine

Gemini pins machines on the floor and pulls specs from manuals and the web.

3 of 3

listed machines placed

›

03 · SIMULATE

AI crew runs a shift

Agents make real decisions, the bottleneck shows up, fixes are tested on a copy.

+36%

output with agents, one run

03 — OPPORTUNITIES

Three toolsets today. One photo with Ada-Twin.

HOW IT IS DIFFERENT

Enterprise digital twins Tecnomatix, Omniverse

Misses: Needs CAD models, LIDAR scans, months of setup

Photo-to-3D tools Polycam, Scaniverse, Luma

Misses: A 3D scene with no machine states or production flow

Traditional simulators AnyLogic, FlexSim

Misses: An abstract 2D graph that needs a simulation expert

Ada-Twin

All three, from one photo: a living 3D plant with machines, flow and an AI crew

HOW IT SOLVES THE PROBLEM

Months → minutes

One panorama, about 12 min measured. No CAD, no laser scans.

Within reach of any plant

About US$2.40 per 3D world, and no simulation team needed.

Zero-risk “what if”

Test floor changes and maintenance on a copy, before risking real downtime.

UNIQUE SELLING POINTS

01 **One photo to walkable 3D**

Real metric scale and a collision mesh.

02 **Auto-discovered twin**

Gemini pins machines, reads manuals, drafts the flow.

03 **An agent crew that explains**

Planner, technician, supervisor, each with a risk persona.

04 **Every number traceable**

Tagged sourced, web-cited or assumed.

Not just a 3D model. Not just a math simulator. An AI-operated virtual factory that tests real decisions before real spending.

04 — LIST OF FEATURES OFFERED BY THE SOLUTION

Everything between a photo and a tested fix.

01 · WORLD LABS MARBLE

Spatial 3D

- 360° photo to a 3D Gaussian-splat world
- Collision mesh and real metric scale
- Browsable viewer with pinned machines

02 · GEMINI VISION

Perception

- Locates machines and pins them in 3D
- Specs from manuals and the web
- AI-drafted material flow, edited by chat

03 · GEMINI + ADK

AI crew

- Planner, technician and supervisor roles
- Cautious, aggressive or by-the-book
- Plain-English reasons; baseline fallback

04 · DISCRETE-EVENT ENGINE

Simulation

- 8-hour shift replayed in about 2 minutes
- Heat, demand, absence, supply stress
- Live-streamed events and agent actions

05 · SIMULATION ENGINE

Analytics

- Bottleneck flagged automatically
- Throughput vs target, scrap and WIP
- Station utilisation and downtime

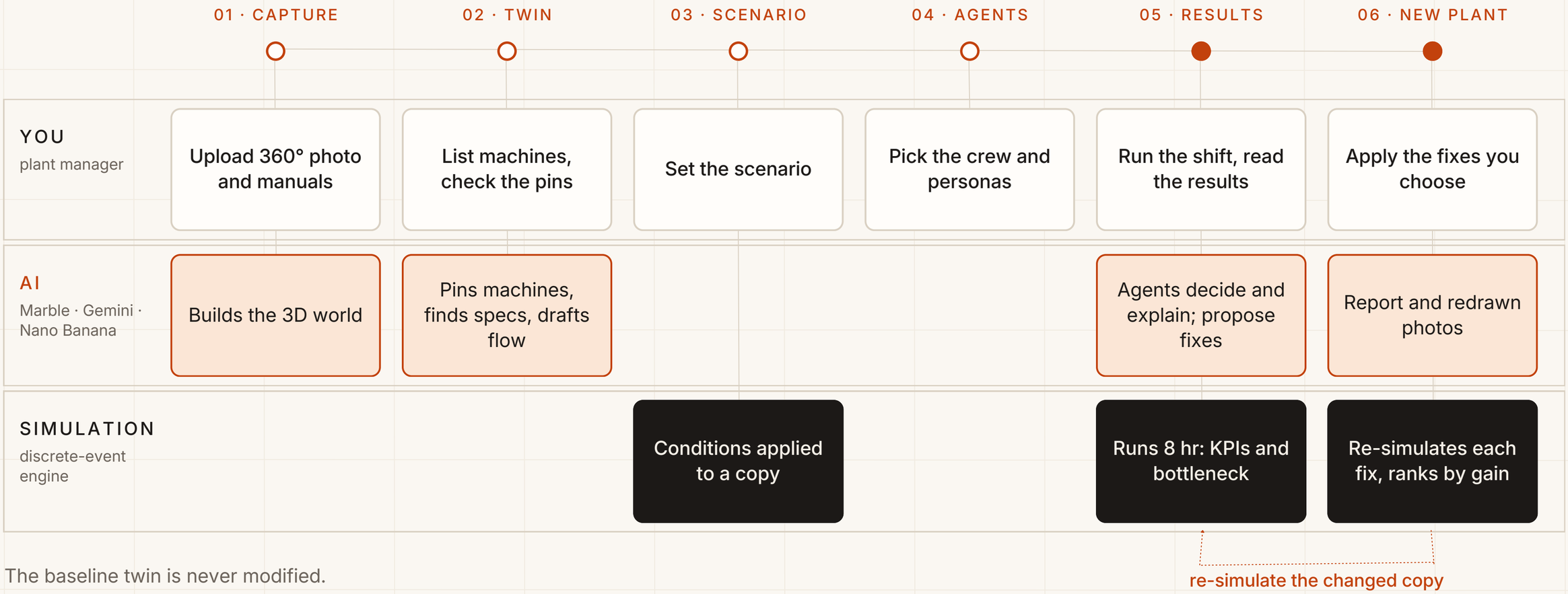
06 · NANO BANANA

Optimisation

- AI-proposed fixes tested on a copy
- Same-seed re-runs, ranked by gain
- Your photos redrawn with the changes

05 — PROCESS FLOW DIAGRAM

From one photo to a tested fix, with a loop.

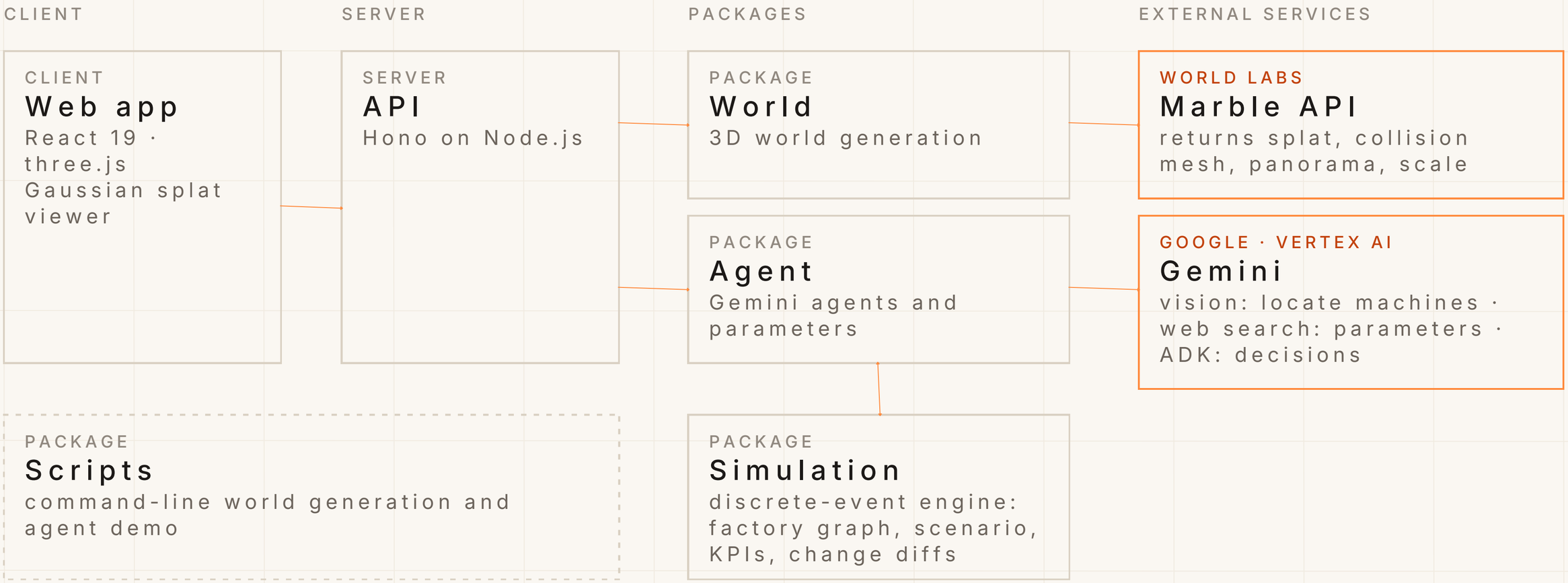


The baseline twin is never modified.

Wireframes/Mock diagrams of the proposed solution (optional)

06 — ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION

A thin web app over three focused packages.



07 — TECHNOLOGIES TO BE USED IN THE SOLUTION

Gemini at the core, on Google Cloud.

AI · VERTEX AI	Gemini on Vertex AI gemini-flash-latest	GenAI SDK @google/genai	ADK Agent Development Kit	Nano Banana Google image model
VISION AI	Gemini vision image analysis: locates each machine in the 360° panorama			
FRONTEND & 3D	React 19	three.js Gaussian splat rendering	Marble 1.1 Plus World Labs	
BACKEND	Hono	Node.js Node 24		
LANGUAGE & BUILD	TypeScript	Turborepo monorepo	pnpm	
DEPLOYMENT	Cloud Run runs the API container	Cloud Build builds the image	Artifact Registry stores images	Docker Compose for local runs
DATA & SECRETS	Cloud SQL for PostgreSQL	Drizzle ORM	Cloud Storage uploaded documents	Secret Manager World Labs API key

08 — ESTIMATED IMPLEMENTATION COST

A 3D twin and simulated shift for less than a cup of coffee.

Costs from live API calls and prototype benchmarks. No GPU servers or special hardware.

COMPONENT	SERVICE AND USAGE	COST
3D world	World Labs Marble 1.1 Plus, 1 panorama	**$1.20–2.40**
Draft preview	Marble draft, 150 credits	**$0.12**
Machine discovery	Gemini vision and web search	**~$0.01**
8-hour shift	Gemini Flash via ADK, ~30 decisions	**~$0.008**
Redesign renders	Nano Banana, 2–3 images	**~$0.08**
Hosting	Cloud Storage, Cloud Run free tier	**<$0.01**
Total per twin run	Photo → twin → 8 h shift → fix	**~$1.50–2.60**

Traditional digital twin

$50,000+ · 3 months

Ada-Twin

~$2.50 · under 15 min

Cheap to scale. Re-running a shift costs under 1 cent.

Zero CapEx. No GPU workstation or CAD licences; runs in-browser.

Draft first. Check the layout for $0.12 before a full run.

09 — SNAPSHOTS OF THE PROTOTYPE



3D splat viewer with collision mesh

• WORKING



Gemini agents operating a shift

• WORKING



Report page, sample data

○ MOCK-UP

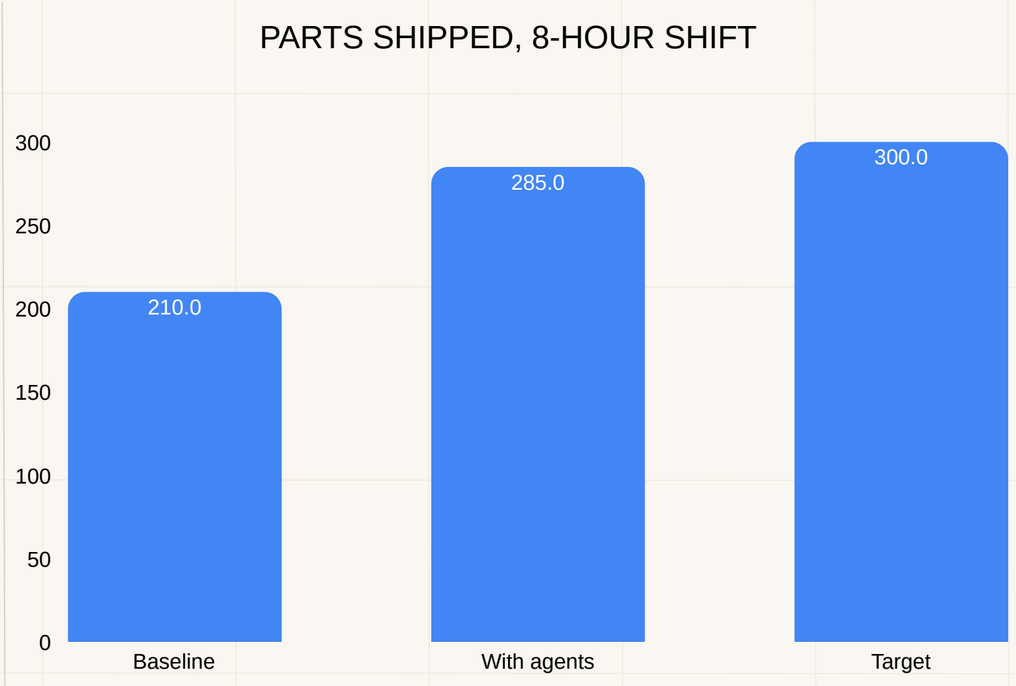
10 — PROTOTYPE PERFORMANCE REPORT / BENCHMARKING

+36% output with Gemini agents. From one run, not an average.

Scenario “Hot week, full order book”: 8-hour shift · 34 °C · one maintenance crew · target 300 parts · same random seed

	BASELINE	WITH AGENTS
Parts shipped	210	285
Throughput per hour	26.3	35.6
Boilers down	33%	8%
Parts scrapped	2	11

Target of 300 missed by 15. Scrap rose from 2 to 11.



11 — PROTOTYPE PERFORMANCE (CONTINUED)

Fast enough to use. Checked where we can.

~12 min

3D world generated from a panorama

~2.5 min

simulated shift with 30 agent decisions

3 of 3

listed machines placed by Gemini; position accuracy to be measured

SIMULATION ENGINE

Deterministic per seed

Its tests check it against queueing theory.

PUMP WITH NO DOCUMENTS · 5 PARAMETERS



■ 2 sourced from the web, cited ☐ 3 marked as assumed

12 — ADDITIONAL DETAILS / FUTURE DEVELOPMENT

Finish the loop. Widen the twin. Prove it at scale.

01

Finish the product

- Build the report and redesign steps for real
- Draft the whole production line from the photo, not by hand

02

Widen the twin

- Multi-room plants by joining worlds
- Phone video scans as an alternative to generated worlds
- Live sensor data

03

Prove and extend

- Averaged multi-seed benchmarks
- Failure-prediction models trained on maintenance logs with Vertex AI
- More agent roles

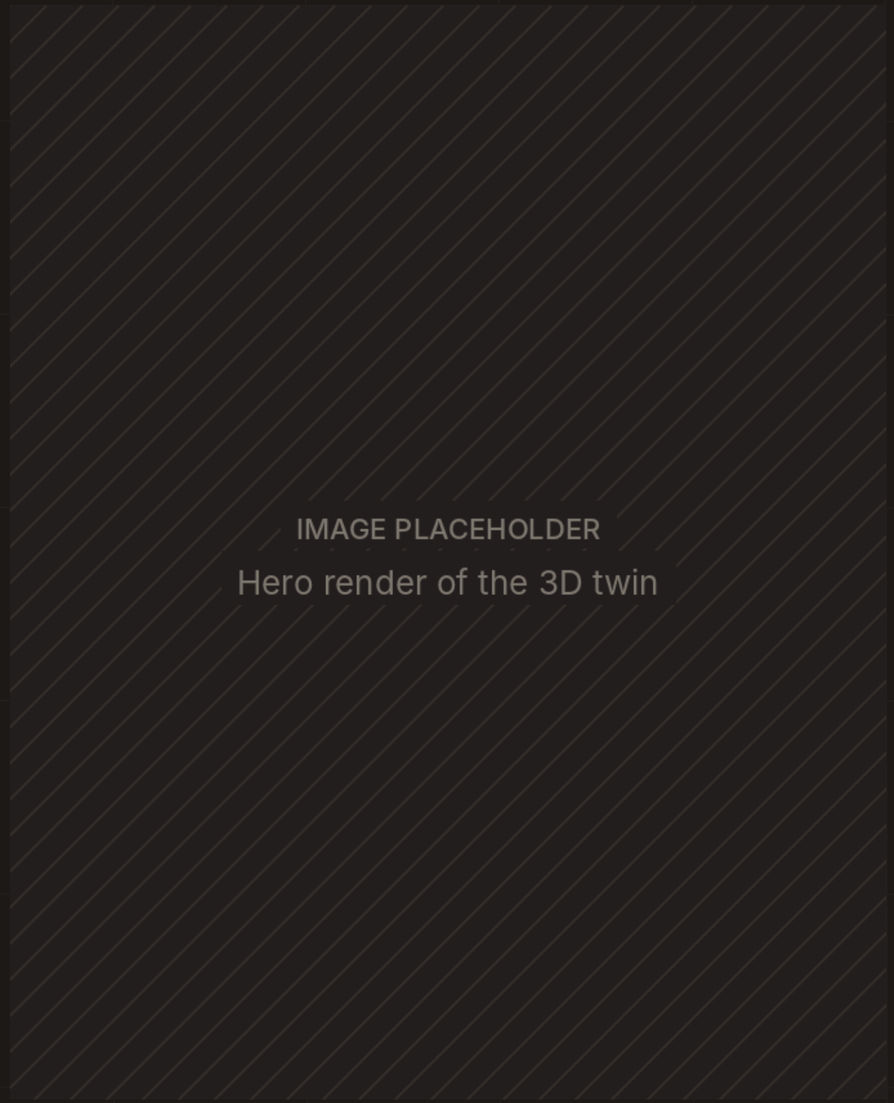
13 — LINKS

Try it yourself.

01	GitHub public repository	[link to be added]
02	Demo video, 3 minutes	[link to be added]
03	Final product	[link to be added]

ADA-TWIN — THE ADAPTIVE FACTORY TWIN

One photo in. A walkable twin out. Fixes tested before the real floor.



OUR ASK

[to be added by the team]

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

